

Course: B. Pharm.

Semester: III

Subject with Subject Code: Pharmaceutical Organic Chemistry-II (BP301T)

Date: 18/05/2018

Marks: 75

Duration: 3hrs

Instructions: i) All questions are compulsory

ii) Figures to the right indicate full marks

iii) Draw the diagrams or flow charts wherever necessary.

Q. 1 Choose the correct alternative.

20

- Which statement of the following gives false information about benzene?
 - It is immiscible with water forming the lower layer.
 - It is a planar molecule with bond angle 120° .
 - It can be converted into cycloheptane by hydrogenation at 200°C in the presence of Ni catalyst.
 - It reacts with ethyl chloride in the presence of aluminium chloride to form ethylbenzene.
- Benzene undergoes substitution reactions more easily than addition reactions because:
 - It has a cyclic structure
 - It has double bonds
 - It has six hydrogen atoms
 - There is delocalization of electron
- Benzene reacts with H_2 at 150°C at 30 atm in presence of Ni catalyst to give:
 - Cyclohexane
 - Cyclohexene
 - n - Hexane
 - None of the above
- The electrophile which is considered to be the active agent in the nitration of benzene is:
 - NO_2
 - NO^+
 - NO_2^+
 - HNO_2^+
- Sodium or potassium salts of fatty acids are called
 - Surfactants
 - Detergents
 - Carbohydrates
 - Soaps
- Partial hydrogenation of vegetable oils in the presence of Ni catalyst at 200°C gives
 - Vanaspati ghee
 - Margarine
 - Both (A) and (B)
 - None of these
- The degree of unsaturation of a fat can be determined by means of its
 - Iodine number
 - Octane number
 - Saponification number
 - Melting point

8. Ozonolysis of naphthalene ring gives
A. Phthalic acid
B. Phthaledehyde
C. Phthalic anhydride
D. Napthaquinone
9. _____ compound is used for preparation of dyes such as; malachite green, bromocresol green.
A. Diphenylamine
B. Anthracene
C. Naphthalene
D. Triphenylamine
10. Anthracene undergoes electrophilic substitution reactions mainly at _____.
A. C-1
B. C-2
C. C-9
D. C1 & C-2
11. Naphthalene undergoes nitration with $\text{HNO}_3/\text{H}_2\text{SO}_4$ at $50-60^\circ\text{C}$ to give mainly _____.
A. 1-nitronaphthalene
B. 2-nitronaphthalene
C. 1,2-dinitronaphthalene
D. 1,8-dinitronaphthalene
12. Cycloalkanes have similar formula as _____.
A. Alkanes
B. Alkenes
C. Alkynes
D. Cycloalkenes
13. Which of the following is treated with sodium in dry ether to give cyclopropane?
A. 1,1-dibromopropane
B. 1,2-dibromopropane
C. 1,3-dibromopropane
D. 2,2-dibromopropane
14. Which of the following cycloalkane is not expected to have ring strain?
A. Cyclobutane
B. Cyclohexane
C. Cyclopropane
D. Cycloheptane
15. The most stable confirmation of cyclohexane is the _____.
A. Haworth
B. Chair
C. Boat
D. Newmann
16. When phenol reacts with neutral FeCl_3 solution it develops _____.
A. Yellow color
B. Orange color

- C. Green color
- D. Violet color

17. Sodium phenoxide reacts with CO_2 at 125°C under 5atm pressure to give salicylic acid.

This reaction is called

- A. Kolbe's reaction
- B. Perkin reaction
- C. Wurtz reaction
- D. HVZ reaction

18. Benzoic acid on heating with soda lime gives _____

- A. Sodium phenoxide
- B. Benzene
- C. Benzaldehyde
- D. Benzophenone

19. Which of the following reagent is used to prepare benzenediazonium chloride from aniline?

- A. $\text{NaNO}_2 + \text{HCl}$
- B. $\text{NH}_2\text{NH}_2 + \text{KOH}$
- C. LiAlH_4
- D. NaOH

20. Which of the following is strongest acid?

- A. Trichloroacetic acid
- B. Phenol
- C. Acetic acid
- D. Benzoic acid

Q. 2 Answer **any two** of the following questions.

20

- A. Explain the Nitration & sulphonation of benzene.
- B. Explain the method of preparation and reactions of Naphthalene.
- C. Define fatty acids. Explain Reichert-Meissl (RM) Value and Saponification Value in detail. Comment on rancidity of oils.

Q. 3 Answer **any seven** of the following questions.

35

- A. Explain the Basicity of Amines.
- B. Give the reactions of Benzoic Acid.
- C. Explain the stability of Cycloalkane.
- D. Explain the chemical reactions of Phenanthrene.
- E. Give structure and uses of
 - a. Benzene hexachloride
 - b. Chloramine-I
- F. Write a note on Friedel Craft's Alkylation.
- G. Explain various methods for preparation of Phenols
- H. Explain the significance and principle involved in determination of Acid value and Iodine value.
- I. Give the methods of preparation and reactions of Cycloalkanes.

*** End ***



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY
LONERE - RAIGAD - 402 103

Winter Semester Examination - December - 2019

Branch: B. Pharm. Second Year

Subject: Pharmaceutical Organic Chemistry-II (BP301T)

Date: 17/12/2019

Semester: III

Marks: 75

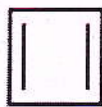
Time: 3hrs

- Instructions: i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

Q. No.1 Multiple Choice Questions (MCQs) = $20 \times 1 = 20$

(Answer all the questions)

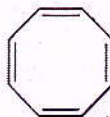
- 1) The Bond angle between carbon atoms in cyclohexane is -----
a) 60° b) 90° c) 120° d) 109.28°
- 2) Anthracene undergoes electrophilic substitution reactions mainly at -----
a) C-1 b) C-2 c) C-9 d) C-1 & C-2
- 3) Phenanthrene on reduction in presence of Na in isopentanol gives -----
a) 9,10 -dihydrophenanthrene
b) 1,2 -dihydrophenanthrene
c) 3,4 -dihydrophenanthrene
d) None of the above
- 4) The most stable conformation of cyclohexane is the -----
a) Haworth b) Chair c) Boat d) Newmann
- 5) Fats and oils are -----
a) Diesters of glycerol b) Diesters of glycol
c) Monoesters of glycerol d) Triesters of glycerol
- 6) Naphthalene undergoes nitration with $\text{HNO}_3/\text{H}_2\text{SO}_4$ at $50-60^\circ\text{C}$ to give mainly -----
a) 1- Nitronaphthalene b) 2- Nitronaphthalene
c) 1, 2 Dinitronaphthalene d) 1, 8- dinitronaphthalene
- 7) Which of the following compound is aromatic



a)



b)



c)



d)

- 8) Liquid oil can be converted in to solid fats by -----
- a) Saponification b) Hydrolysis
- c) Hydrogenation d) Oxidation of double bond
- 9) Phenol on reaction with excess of Bromine water to give -----
- a) bromobenzene b) m-bromophenol
- c) ortho and p-bromophenol d) 2,4,6-tribromophenol
- 10) Which of the following acid is weaker than Benzoic acid?
- a) p-nitrobenzoic acid b) p-methylbenzoic acid
- c) p-chlorobenzoic acid d) o-chlorobenzoic acid
- 11) The degree of unsaturation of fat can be determined by means of its -----
- a) Iodine number b) Saponification number
- c) Acetyl number d) Ester number
- 12) The carbon atoms in a benzene ring are -----
- a) sp^3 hybridised b) sp hybridized c) sp^2 hybridised d) None of the above
- 13) Phenols can be synthesized by the ----- methods
- a) Dow's process b) From cumene c) From coal tar d) All of the above
- 14) Benzene undergoes substitution reaction more easily than addition reaction because:
- a) It has a cyclic structure b) It has double bonds
- c) It has six hydrogen atoms d) There is delocalization of electrons
- 15) Highly unsaturated oil exposed to air, they undergo oxidation and polymerization to form----
- a) Drying oil b) Rancid oil c) Hardening oil d) Saponification
- 16) Anthracene on oxidation gives -----.
- a) Phthalic acid b) Benzophenone c) Benzoic acid d) Anthraquinone
- 17) Benzene reacts with acetic anhydride in presence of $AlCl_3$ to form:
- a) Benzophenone b) Acetophenone c) Phenylacetate d) Chlorobenzene
- 18) Aniline on reaction with acetic anhydride gives -----.

a) N-methylaniline

b) p-aminoacetophenone

c) Acetanilide

d) m-aminoacetophenone

19) Which of the following compound is more reactive?

a) Cyclohexane

b) Cyclopentane

c) Cyclopropane

d) Cycloheptane

20) Benzene reacts with Conc. HNO_3 in presence of Conc. H_2SO_4 to give nitrobenzene, this is an example of ----.

a) Electrophilic substitution

b) Electrophilic addition

c) Nucleophilic substitution

d) Nucleophilic addition

Q. No.2 Long Answers (Answer 2 out of 3) = $2 \times 10 = 20$

A] Discuss Aromaticity and Resonance in Benzene. Explain any two Electrophilic Substitution Reactions with Example.

B] Give Synthesis, Reactions and Medicinal uses of Phenanthrene and Anthracene.

C] Explain Principle, Procedure involved in Determination of Saponification Value. Give Significance of Reichert Meissl (RM) value.

Q. No.3 Short Answers (Answer 7 out of 9) = $7 \times 5 = 35$

A] Explain the Structure and Uses of DDT and Saccharine.

B] Discuss Baeyer strain Theory of Stability of cycloalkanes.

C] Give the structure and synthesis of triphenylmethane.

D] What are fats and oil? Give significance of iodine value and ester value.

E] Explain the effect of substituent on the acidity of aromatic acids/benzoic acid.

F] What are aryl diazonium salts and give its applications.

G] Give qualitative tests for phenols, give structure and uses of cresol and resorcinol.

H] Give halogenations reaction and reamer-tiemann reaction for phenol.

I] Explain Friedel-Crafts alkylation and acylation reaction of benzene.

— END OF PAPER —

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 29/12/2022

Course : B. Pharmacy

Subject Name : Pharmaceutical Organic Chemistry-II

Max Marks : 75

Sem: III

Subject Code: BP301T

Duration : 3 Hr.

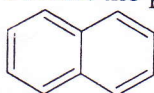
Instructions:

1. All questions are compulsory
2. Draw Structures/ diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Give Structure and uses of DDT.
- ii) Define Acid Value and give its significance.
- iii) Write Structure and uses of Resorcinol.
- iv) Write down qualitative test for phenol.
- v) Write Huckel rule for aromaticity with suitable example.
- vi) Predict the product



H_2/Pt

?

- vii) Give any two reactions of benzoic acid.
- viii) Discuss structure and medicinal uses of Naphthalene.
- ix) Write any two methods for preparation of diphenylmethane.
- x) Write down the structure and give numbering to the derivatives of naphthalene and Anthracene

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

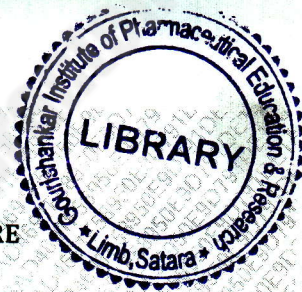
- i) Define fats and oils with example and give difference between them. Explain Hydrolysis, Hydrogenation and Rancidity reactions of fats.
- ii) What are condensed polynuclear hydrocarbon. Write down Haworth synthesis of Anthracene and any two-chemical reaction of it. Draw the structure of derivatives of it along with uses.
- iii) What is Friedel craft reaction. Explain in detail Friedel Crafts alkylation and acylation. Give its limitations.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) What are activating & deactivating groups?
- ii) Explain reaction and mechanism of nitration, halogenation and sulphonation in benzene.
- iii) Write down about effect of substituents on acidity of phenol. Give synthetic uses of aryl diazonium salt.
- iv) Give analytical and synthetic evidences in the derivation of structure of benzene.
- v) Explain electrophilic substitution in diphenylmethane. Give its uses.
- vi) Write Electrophilic substitution reaction of phenanthrene.
- vii) Discuss Baeyer's Strain theory.
- viii) Give reactions of cyclopropane and cyclobutane.
- ix) Explain the basicity of aromatic amine

-----END OF THE PAPER-----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary End Semester Examination – Summer 2022

Course: B. Pharmacy

Sem: III

Subject Name: Pharmaceutical Organic Chemistry-II

Subject Code: BP301T

Max Marks: 75

Date: 24/08/2022

Duration: 3.45 Hrs.

Instructions –

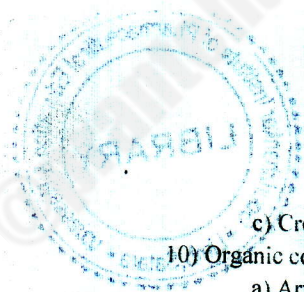
1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

I. Multiple Choice Questions (All questions are compulsory)

(20 Marks)

- 1) Benzene reacts with acetic anhydride in presence of $AlCl_3$ to form
 - a) Benzophenone
 - b) Acetophenone
 - c) Phenyl Acetate
 - d) Chlorobenzene
- 2) Which of the following cycloalkane has the lowest heat of combustion per carbon item?
 - a) Cyclopropane
 - b) Cyclopentane
 - c) Cyclohexane
 - d) None of these
- 3) Acetylenes Contain
 - a) Carbon – Carbon Bond (c-c)
 - b) Carbon – Carbon Double Bond
 - c) Carbon – Carbon Triple Bond
 - d) No Bond.
- 4) Which Catalyst used during halogenation Benzene
 - a) Lewis Acid
 - b) Lewis Base
 - c) Platinum
 - d) Ni/ Platinum
- 5) Which of the following is used in manufacturing of Bakelite
 - a) Phenol
 - b) Formaldehyde
 - c) Ethyl Alcohol
 - d) Both A and B
- 6) Which of the following is most acidic
 - a) Benzyl alcohol
 - b) Para Chloro phenol
 - c) Para Fluoro Phenol
 - d) Meta Chloro Phenol
- 7) Which of the following statement is true regarding Halogenation of Benzene
 - I. Activation of electrophile by Lewis catalyst
 - II. Attack of electrophile by aromatic ring
 - III. Attack of aromatic ring by activated electrophile
 - IV. Activated electrophile protonation to regenerate aromatic ring
 - a) I & III
 - b) I & IV
 - c) I & II
 - d) II & IV
- 8) Halogen through are electron withdrawing in nature they are ortho, para directors in electrophilic aromatic substitution.
 - a) Positive charge accommodated on halogen
 - b) They are ring activators
 - c) Formation of stable hydronium ion
 - d) Because they are electronegative in nature
- 9) Which is most acidic in following compounds:
 - a) 2 chloro-propanol
 - b) m-chlorophenol

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- c) Cresol
10) Organic compound which contains one or more Benzene ring are termed as
a) Aryl
c) Benzenes
d) Phenol
b) Arenes
d) Benzyls
- 11) Benzimidazolium chloride reacts with _____ to yield benzene
a) H_3PO_2
c) KCN
b) HBF_4
d) SnO_2
- 12) Phenols with _____ gives colored complexes
a) Diazonium Salt
c) $FeCl_3$
b) Bromine
d) Aspirin
- 13) Mono Nitration of n-methyl aniline gives _____ used as a major product
a) Ortho-nitro, n-methyl aniline
c) Meta-nitro, n-methyl aniline
b) Para-nitro, n-methyl aniline
d) Ortho, Para di-nitro, n-methyl aniline
- 14) Sulphonation of phenol at room temperature gives
a) *p*- isomer
c) *m*-isomer
b) *o*- isomer
d) Both *o*- and *p*- isomers
- 15) The degree of unsaturation of lipid can be measured as
a) Saponification
c) RM
b) Iodine
d) Polenske number
- 16) Molecular formula for DDT is
a) $C_{14}H_9Cl_5$
c) $C_{10}H_9Cl_5$
b) $C_{10}H_7Cl_5$
d) $C_{10}H_9Cl_2$
- 17) Which of the following is weaker than benzoic acid
a) *p*- Nitrobenzoic Acid
c) *p*-Methylbenzoic Acid
b) *p*- Chlorobenzoic Acid
d) *o*- Chlorobenzoic Acid
- 18) Highly unsaturated oil exposed to undergo oxidation and polymerization to form
a) Drying Oil
c) Rancid Oil
b) Hardening Oil
d) Saponification
- 19) Which of the following is not ortho-para directing
a) -OH
c) -NH₂
b) -SH
d) -CN
- 20) Which of this is the simplest example of polynuclear hydrocarbon
a) Pyrene
c) Naphthalene
b) Dibenzoanthracene
d) None of these

II. Long Answers (Answer any Two)

(20 Marks)

- 1) Write the principle, reaction and mechanism of Nitration and Friedel Craft's Alkylation in benzene.
- 2) Give the synthesis, reactions and medicinal uses of anthracene & naphthalene
- 3) How acidity of aromatic carboxylic acid is analyzed. Explain the effects of substituents on acidic strength of benzoic acid.

III. Short Answers (Answer any Seven)

(35 Marks)

- 1) Write the methods of preparations of diphenyl methane.
- 2) Write the mechanism for diazotization reaction.
- 3) Explain the Huckel's rule with suitable examples.
- 4) Define angle strain. Discuss why higher cycloalkanes are more stable than lower members.
- 5) Write the chemical reactions of fats and oils.

- 6) Explain the molecular orbital structure of benzene.
- 7) Explain the directive effects of substituent towards electrophilic substitution reactions of benzene.
- 8) Give the structure and used of saccharin, resorcinol and phenanthrene.
- 9) Explain Bayer's strain theory and Sacht-Mohr's theory.

-----END OF PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 22/07/2023

Course : B. Pharmacy

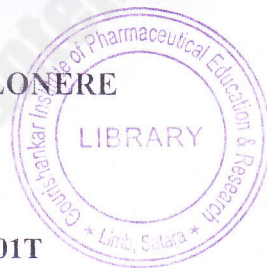
Subject Name : Pharmaceutical Organic Chemistry-II

Max Marks : 75

Sem: III

Subject Code: BP301T

Duration : 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Structure and uses of DDT.
- ii) Define Reichert Meissl Value and give its significance.
- iii) Structure and uses of Resorcinol.
- iv) Write down qualitative test for phenol.
- v) Huckle rule for aromaticity.
- vi) Saponification of oils.
- vii) Any two reactions of benzoic acid.
- viii) Structure and medicinal uses of Naphthalene.
- ix) Any two method for preparation of diphenylmethane.
- x) Coulson and Moffitt's modification.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Define fats and oils with example and give difference between them. Explain acid value, and saponification value with significance.
- ii) What are condensed polynuclear hydrocarbon. Write down Haworth synthesis of Anthracene and any two-chemical reaction of it. Draw the structure of derivatives of it along with uses.
- iii) Explain in detail Friedel Crafts alkylation and acylation in benzene with mechanism of reaction. Give its limitations.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Explain in detail evidences given by Kekule to establish the structure of benzene.
- ii) Explain reaction and mechanism of nitration, halogenation and sulphonation in benzene.
- iii) Write down about effect of substituents on acidity of phenol. Give synthetic uses of aryl diazonium salt.
- iv) Explain in detail hydrogenation of oils. Give preventive major for rancidity of oils.
- v) Explain electrophilic substitution in diphenylmethane. Give its uses.
- vi) Give any three reactions of Phenanthrene. Note on derivatives of it.
- vii) Discuss Baeyer's Strain theory.
- viii) Give reactions of cyclopropane and cyclobutane.
- ix) Define ester value give significance and principle involved in its determination.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2023

Date: 02/01/2024

Course : B. Pharmacy

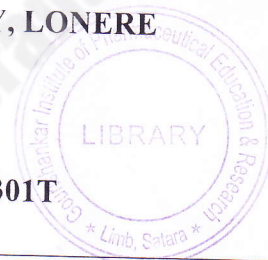
Subject Name: Pharmaceutical Organic Chemistry-II

Max Marks : 75

Sem: III

Subject Code: BP301T

Duration : 3Hr.



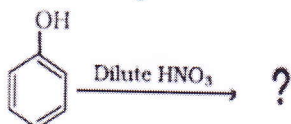
Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Define saponification value. Give its significance.
- ii) Draw structure and give medicinal uses of naphthalene.
- iii) Write synthetic uses of aryl diazonium salts.
- iv) Predict the product:



- v) Discuss limitations of Friedel crafts alkylation.
- vi) Draw structure and write uses of BHC.
- vii) Give reason: Though halogens are o,p-directors, they are ring deactivators.
- viii) Draw structures of derivatives of anthracene and phenanthrene.
- ix) Write Huckel's rule of aromaticity with suitable example.
- x) Write any two reactions of cyclopropane.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) What are phenols? Explain acidity of phenols. Write any three methods of preparation and three reactions of phenols.
- ii) Enlist analytical constants of oils and fats. Discuss in detail about iodine value and Reichert-Meissl value and give their significance.
- iii) Discuss analytical, synthetic and other evidences in the derivation of structure of benzene.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Discuss Baeyer's strain theory with its limitations.
- ii) Explain hydrogenation and saponification reactions of fats and oils.
- iii) Write ring substitution reactions of aromatic amines. Discuss effect of substituents on basicity of aromatic amines.
- iv) Draw resonance structures of phenol and aniline.
- v) Define aromatic electrophilic substitution reaction. Explain mechanism of nitration and sulphonation of benzene.
- vi) Write methods of synthesis and reactions of cycloalkanes.
- vii) Give reason. Naphthalene undergoes electrophilic substitution reactions preferentially at α position.
- viii) Give any two methods of preparation and reactions of benzoic acid.
- ix) Explain structure, synthesis and medicinal uses of Diphenylmethane.

-----END OF THE PAPER-----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Pharmacy

Subject Name: Pharmaceutical Organic Chemistry- II

Max Marks: 75

Date: 12-06-2024

Semester: III

Subject Code: BP301T

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions)

(10X2) =20

- i) Define iodine value. Give its significance.
- ii) Draw the structure and general uses of Saccharin.
- iii) Enlist the analytical constants of fats and oils.
- iv) Write the structure and medicinal uses of Anthracene.
- v) Draw the structure of resorcinol and 2-naphthol.
- vi) Enlist four ortho, para and meta director from each class.
- vii) What is Aromaticity? Why benzene is called as an aromatic compound?
- viii) Draw the structure of BHC and Chloramine.
- ix) What is Saponification value and give its significances.
- x) What are some limitations of baeyer's strain theory?

Q.2. Long Answers (Answer 2 out of 3)

(10X2) =20

- i) What are polynuclear aromatic hydrocarbons? Give the synthesis reactions of naphthalene, Phenanthrene.
- ii) Explain nitration, sulphonation and halogenation of benzene with examples and Mechanism.
- iii) Explain the principle & procedure involved in the determination of acid value of oil Sample.

Q.3. Short Answers (Answer 7 out of 9)

(5 X 7) = 35

- i) Despite being ortho and para director, chlorine exhibit deactivating properties within a molecule.
- ii) Explain the facts supporting kekule structure of Benzene.
- iii) Explain the effect of substituents on acidity of phenols.
- iv) Discuss the method of synthesis of cyclopropane and cyclobutane.
- v) Explain the Huckel's rule for naphthalene and Anthracene.
- vi) Discuss the basicity of aromatic amines.
- vii) What is the diazonium reaction? Explain the general reaction?
- viii) Add a note on the chemical reaction of Anthracene.
- ix) Discuss Bayer's strain theory with suitable examples.

*** END OF THE PAPER ***